Introduction of immunology

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Lecture 1:

Contents

- Objective
- Definition of immunity
- Branches of immunology
- Application of immunology
- Terms in immunology
- Cells of immune system

Objective

- The student must be learn definition , branches and application of immunology. Also there are many immunological terms should be learned by students.

Definitions of immunology:-

Branch of medical science that deals with the study of development and function of both **cellular and humeral** components of the immune system by which the body reacts to expel, destroy or neutralize foreign substances including pathogenic microorganisms.

Immunity:-

- Refers to all the physiological mechanisms that allow the body with the capacity to recognize materials as **foreign** to it and to **neutralize, eliminate or metabolize** them.
- Although the usual outcome of immune response is **beneficial**, it may cause **harmful** effects such as autoimmunity and hypersensitivity.
- **Immunis i**s a Latin word meaning free from burden or taxes.

Branches of immunology:-

- **1-Immunochemistry:** Deals with the chemical nature of **antigen, antibody** and their interactions.
- **2-Immunobiology**: Related to biologic functions of the **cells and tissue** component of the immune system.
- **3-Immunogenetics:** Study of **immunoglobulin genes** and the genetic basis of immune response.

Branches of immunology:-

- **4-Clinical immunology:** Deals with pathogenesis, diagnosis and treatment of **immune mediated** diseases.
- **5-Immunotherapy:** Deals with prevention and treatment of diseases by **immunological meanse** as vaccination, immunoglobulin therapy and desensitization.

Branches of immunology:-

- **6-Immunohematology:** Deals with immunological basis involved in different **hematological disorders**
- **7-Immunopharmacology:** Deals with the pharmacological action of different **secreted molecules** in immune responses.
- **8-Serology:** Study of **immune reactions** mediated by **antibodies or immunoglobulins** present in the serum.

Application of immunology

- 1-Diagnosis of diseases: By **detecting antibody or antigen** it help in diagnosis of many diseases.
- 2-Understanding disease process: Immunology can **explain the pathogenesis** of infectious, autoimmune, neoplastic and many other diseases.

Application of immunology

- 3-Prevention and treatment of diseases: **Immunization** against many diseases by vaccine and immunoglobulins also in **management** of cancer in future.
- 4-Blood transfusion serology: Grouping, typing and cross matching in Blood transfusion.
- 5-Tissue typing and histocompatibility testing necessary for successful **organ transplantation**.
- 6-Forensic medicine: such as **paternity testing**.

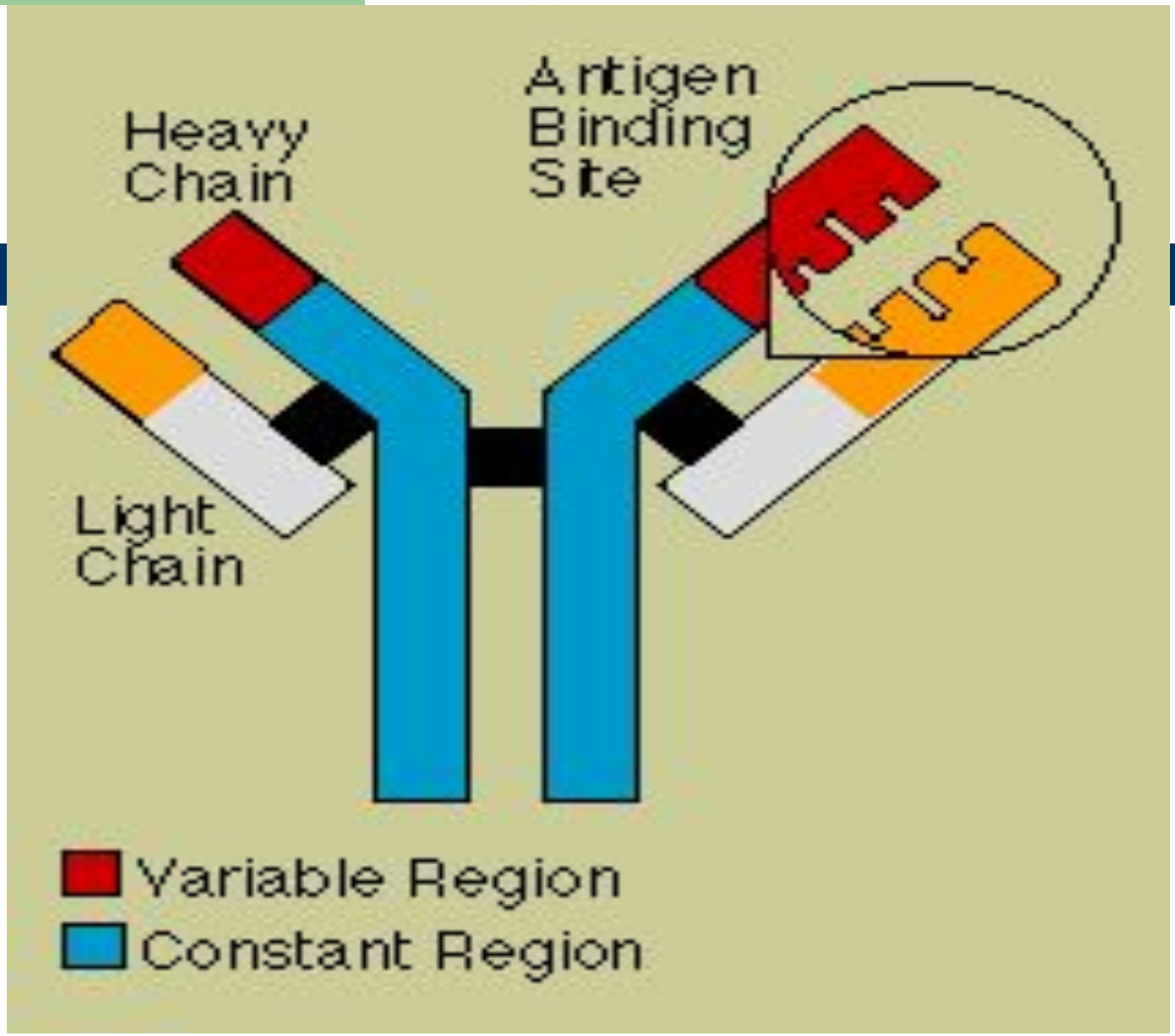
Immune components: are divided into:-

- **1- Humeral factor:** (Immunoglobulins, Complement system, Cytokines).

- **2- Cellular factor:** (Lymphocytes, phagocytes)

Common terms in immunology

- **Antigen:-**
- It is a foreign substance when inter to the body may cause **stimulation** of immune system.
- **Immunoglobulins(Antibodies):-**
- Are glycoprotein that bind **specifically** to the **antigen** that induced their formation, they produced from **plasma cells** as a result of **activation of B –cells** through acquired immune response. The types of immunoglobulins(Igs) are IgM, IgG, IgA, IgD and IgE.


Heavy Chain
Antigen Binding Site
Light Chain
Variable Region
Constant Region

The complement system:-

- Is a group of serum proteins consist of 9 component C1-C9, present in **low and in an active form**, but can be activated to form an **enzyme cascade** leading to **destruction** the target substance.
- Activation of complement component occurs by **classical pathway alternative pathway and lectin pathway.**

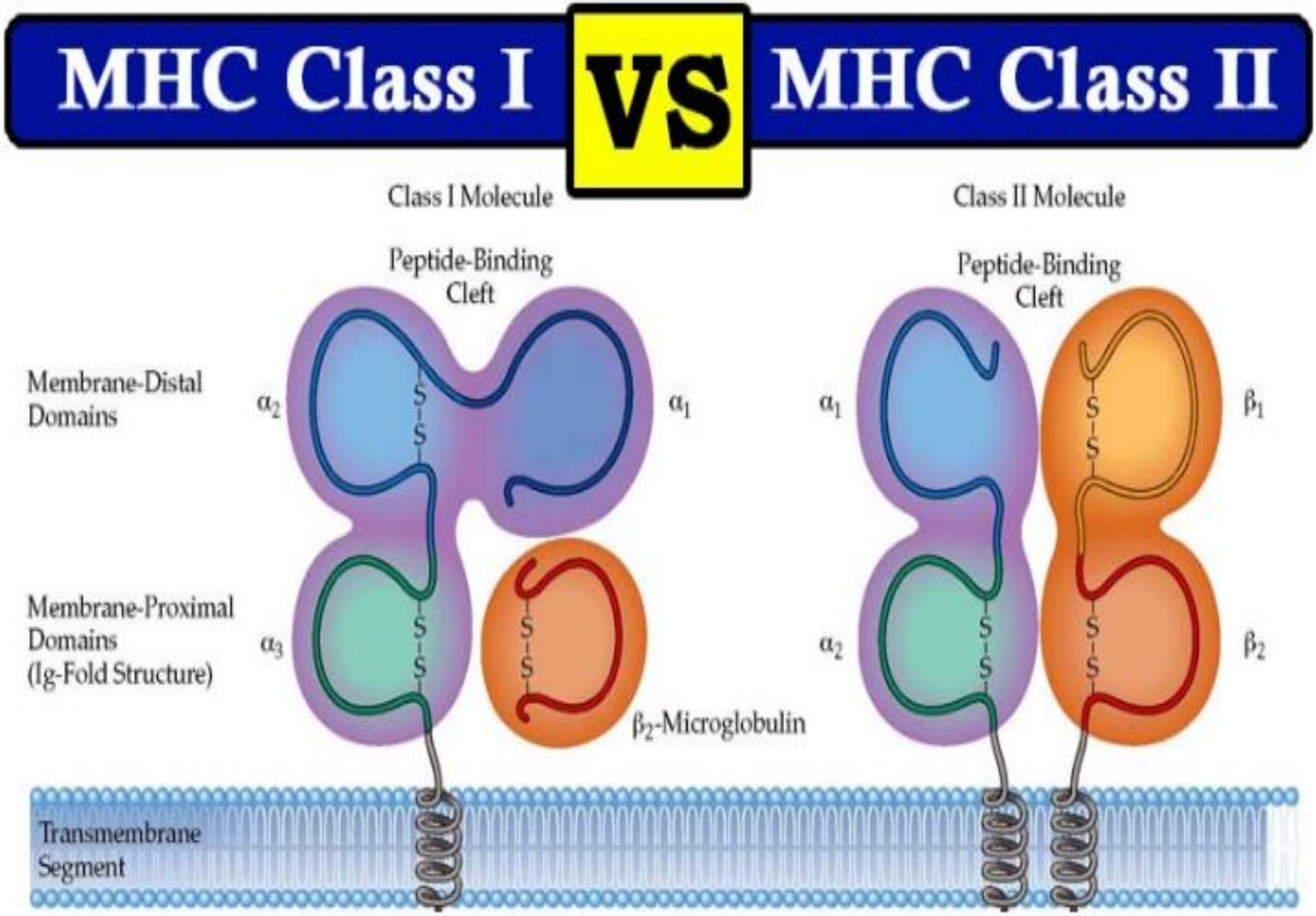
Cytokines:-

- Are **low molecular weight soluble glycoproteins.**
- They are **immunomediators** produced after **stimulation** of immune system play important role in **regulation and orchestrate** the immune system and immune respone.
- Examples on cytokines are lymphokines, monokines, adipokine, chemokine and interleukins.

Major histocompatibility complex (MHC):-

- Cell surface proteins or **antigens** that coded by a cluster of genes present on chromosome number 6, they are **polymorphic** within a species and responsible for **graft rejection** between members of the same species which differ at these loci.
- In human the **MHC called HLA** and classified into 3 classes they are class 1, class II and class III.

Differences between MHC Class I and Class II



Cluster of differentiation (CD)

- The **cluster of differentiation** (**cluster of designation** or **classification determinant** is **cell surface glycoprotein** used for the **identification and investigation** of cell surface molecules
- Acting as receptors or ligands, cell signaling, cell adhesion.
- CD for humans is numbered up to 371 till 2016

Cells of immune system can be classified into:-

- **1-Myloied linage**
- **2- Lymphoid linage:-**
- **The myeloid linage include:-**
- **A-Mononuclear phagocytes**: They are found in the blood and in tissues. In the blood they are known as **monocytes** while in the tissues they differentiate into different histologic forms for example
- 1-In connective tissue they are known as histocytes
  2-In the kidney as measangial cells

- 3- In bone as osteoclasts
- 4-In brain as microglia
- 5- In the spleen, lymph node and thymus as the sinus lining macrophage
- 6- In the liver as Kupffer cells
- 7- In the lungs as alveolar macrophage
- 8-Peritonial macrophage, free floating in peritoneal fluid

B-Multinuclear phagocyte:

- Polymorphonuclear cells(PMN) are include Neutrophil, eosinophil and basophile they kill the target by increase phagocidal activity mechanisms.

The lymphoid linage include:-

A-Small a granular lymphocyte

- 1-B-lymphocyte (B-cell)
- 2- T-lymphocyte (T-cell), sub classified into
- A: T-helper (Th-cell)
- B: T-cytotoxic (Tc-cell)
- C: T-suppressor (Ts-cell)
- **B-Large granular lymphocyte**
- **Natural killer cells (NK cells):-**NK cells are large granular lymphocytes, they play a role in the early stages of viral infection and tumorogenesis.

Organs of the immune system:-

- Primary lymphoid organs.
- Secondary lymphoid organs.

Summary

- **Immunity** refers to all those physiological mechanisms that used by the body to resist agents that are foreign to it.
- There are many branches of immunology .
- There are many application for immunology.
- Definition to many terms in immunology.

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Thanks a lot